A Project Report on

# REGIME-AWARE MULTI-ASSET VOLATILITY FORECASTING USING TEMPORAL FUSION TRANSFORMERS

Submitted in partial fulfillment of the requirements for the degree of

**MASTER OF TECHNOLOGY**

**in**

**AI and Data Science Engineering**

by

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March-2026

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This is to certify that **Abhishek Deep** has successfully completed the project work entitled "**Regime-Aware Multi-Asset Volatility Forecasting using Temporal Fusion Transformers**" in partial fulfillment for the award of **Master of Technology** in **AI and Data Science Engineering** during the year **2025-2026**.

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# *Acknowledgement*

I would like to thank **Dr Rev Fr Joseph C C**, Vice Chancellor, CHRIST (Deemed to be University), **Dr Rev Fr Viju P D**, Pro Vice Chancellor, **Dr Fr Jiby Jose**, Director, School of Engineering and Technology, **Fr Shijin P J**, Assistant Director, School of Engineering and Technology, CHRIST (Deemed to be University), **Dr Raghunandan Kumar R**, Dean, and **Dr E A Mary Anita** , Associate Dean, for their kind patronage.

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# Declaration

I, hereby declare that the project titled "**Regime-Aware Multi-Asset Volatility Forecasting using Temporal Fusion Transformers**" is a record of original project work undertaken for the award of the degree of **Master of Technology** in **AI and Data Science Engineering**. I have completed this study under the supervision of **Dr. Alpha Vijayan**, Department of AI and Data Science Engineering and **Dr. Michael Moses T**, Department of AI and Data Science Engineering.

I also declare that this project report has not been submitted for the award of any degree, diploma, associate ship, fellowship or other title anywhere else. It has not been sent for any publication or presentation purpose.

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# *Abstract*

Predicting how rapidly stock prices will swing tomorrow is one of the trickiest challenges that financial practitioners and academic researchers face, especially in emerging economies where policy shifts and global capital flows frequently disrupt local markets. This project tackles the problem head-on for the Indian stock market by building a forecasting framework that we call the Regime-Aware Temporal Fusion Transformer (RA-TFT). The core idea behind this framework is straightforward: different market conditions—calm stretches, moderately choppy periods, sustained trending phases, and full-blown crises—require different modelling responses, so we explicitly detect these states first and then let the deep-learning model condition its forecasts on whatever state the market happens to be in.

We gathered roughly 2.65 million five-minute candlestick observations covering fourteen blue-chip stocks listed on the National Stock Exchange (NSE) of India over a span of nearly ten years. From this raw price and volume data we engineered a set of 47 features that draw on multiple well-established volatility measures: realized volatility at several time scales, lagged values motivated by the Heterogeneous Autoregressive model, jump statistics derived from bipower variation, range-based variance estimators according to Parkinson, Garman-Klass and Rogers-Satchell, GARCH(1,1) conditional variance, cross-asset correlations inspired by the Dynamic Conditional Correlation framework, and Hidden Markov Model regime labels. All these features feed into the Temporal Fusion Transformer architecture, which offers built-in interpretability through its Variable Selection Network and multi-head attention mechanism.

A carefully designed ablation study separates autoregressive persistence from genuine feature-driven prediction. Model A, which receives the lagged target as an input, achieves a coefficient of determination $R^2$ of 0.976 with a mean absolute percentage error (MAPE) of 1.90 percent. These figures closely match the Heterogeneous Autoregressive model of Realized Volatility (HAR-RV) benchmark ($R^2 = 0.983$), confirming that the TFT can learn the autoregressive structure of volatility as effectively as the purpose-built classical model while additionally offering multi-stock generalization and built-in interpretability. Model B, which deliberately excludes the lagged target, yields $R^2 = 0.258$, establishing that the forty-seven engineered features carry modest but genuine predictive information

beyond simple persistence. Regime-conditioned evaluation reveals strong forecasting accuracy in low, normal, and high volatility states ($R^2 > 0.98$) but noticeable degradation during extreme events ($R^2 = 0.765$), highlighting the difficulty of predicting volatility during market dislocations. The Variable Selection Network independently rediscovers that lagged daily realized volatility dominates all other features by roughly two orders of magnitude, effectively learning the HAR-RV structure without being told about it.

***Keywords*:** Volatility Forecasting, Temporal Fusion Transformer, Hidden Markov Model, GARCH, Indian Stock Market, Deep Learning, Regime Detection, NSE

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# GLOSSARY

| Term | Description |
|---|---|
| **RV** | Realized Volatility: sum of squared intraday log-returns |
| **HAR-RV** | Heterogeneous AutoRegressive model of Realized Volatility |
| **GARCH** | Generalized AutoRegressive Conditional Heteroskedasticity |
| **TFT** | Temporal Fusion Transformer |
| **HMM** | Hidden Markov Model |
| **VSN** | Variable Selection Network |
| **GRN** | Gated Residual Network |
| **LSTM** | Long Short-Term Memory network |
| **NSE** | National Stock Exchange of India |
| **NIFTY** | NSE Fifty: benchmark index of 50 large NSE equities |
| **OHLCV** | Open, High, Low, Close, Volume bar data |
| **BPV** | BiPower Variation: jump-robust variance estimator |
| **RMSE** | Root Mean Square Error |
| **MAPE** | Mean Absolute Percentage Error |
| **DA** | Directional Accuracy |
| **QLIKE** | Quasi-Likelihood loss for volatility estimation |
| **RBI** | Reserve Bank of India |

# Chapter 1

# INTRODUCTION

## 1.1 Background and Motivation

Anyone who has spent time watching stock tickers knows that prices do not move smoothly. Some days the market barely twitches; on others a single news headline sends it into a tailspin. The statistical name for these swings is *volatility*, and getting a handle on it matters enormously in practice. Portfolio managers size their bets according to how wild they expect markets to be. Options traders cannot price a contract without an estimate of future volatility. Banks, too, lean on these numbers—Basel III regulations tie minimum capital buffers directly to volatility-based risk metrics.

For a long time, the dominant toolkit came from econometrics. Engle introduced the ARCH model in 1982, showing that variance itself could be modelled as a time-varying quantity rather than a fixed backdrop. Bollerslev generalised the idea four years later with GARCH(1,1), which became something of an industry workhorse. Extensions followed quickly: Nelson's EGARCH modelled log-variance to sidestep negativity constraints, and GJR-GARCH added an asymmetry term to capture the well-known leverage effect where bad news jolts markets more than equally large good news.

A separate line of work took a different route entirely. Instead of fitting a latent-variance process to daily close-to-close returns, researchers found that simply adding up squared five-minute returns over the trading day gives a directly observable measure called *realized volatility*. Corsi built on this in 2009 with the

HAR-RV model—a surprisingly simple regression on daily, weekly, and monthly RV lags that outperforms GARCH almost universally. The intuition is elegant: scalpers, swing traders, and institutional portfolio rebalancers each operate on different horizons, and the superposition of their activity creates the long-memory signature that HAR-RV captures with just three coefficients. Even today, beating HAR-RV remains a tall order.

Over the past five or six years, deep learning has started to make headway. LSTMs trained on the same inputs as GARCH models showed lower out-of-sample error, particularly during the COVID-19 crash when parametric distributional assumptions fell apart. Attention-augmented recurrent networks and full Transformer architectures pushed the envelope further. But there was always a catch: most of these models work as black boxes. A risk manager can look at a GARCH equation and point to the $\alpha$ and $\beta$ coefficients; with an LSTM there is no easy equivalent. The Temporal Fusion Transformer, or TFT, proposed by Lim and colleagues in 2021, was designed from the ground up to address this gap. Its Variable Selection Networks produce feature-level importance scores at every time step, and its multi-head attention mechanism reveals which historical observations the model relies on most. For the first time, a deep learning architecture offered both competitive accuracy and a credible interpretability story.

## 1.2 Problem Statement

Despite the progress outlined above, several practical gaps remain. First, controlled experiments that separate what the autoregressive lag contributes from what the remaining features contribute are almost nonexistent in the deep learning volatility literature. This matters more than it might seem: consecutive daily RV values share 98.7% of their constituent five-minute bars, so a model that simply parrots yesterday's number already gets $R^2 > 0.97$. Without stripping out that lag, there is no honest way to tell whether the dozens of other hand-crafted features actually help.

Second, the TFT has not been tested in a structured way on Indian equity data. India's market has its own set of quirks—RBI monetary policy announcements that can whipsaw bank stocks, foreign institutional investor flows that shift sectoral

dynamics, and occasional structural jolts like demonetisation—and results from US or European settings do not necessarily carry over.

Third, Hidden Markov Model regime labels, which nicely partition market conditions into calm, moderate, elevated, and crisis states, are seldom fed as live conditioning inputs into a forecasting model. More commonly they are used after the fact for portfolio construction.

Fourth, most studies report only aggregate performance numbers, hiding potentially large differences across sectors and individual stocks.

## 1.3 Objectives

The project set out to accomplish five things:

1. Collect, clean, and preprocess roughly 2.65 million five-minute OHLCV bars for fourteen liquid NSE stocks spanning a decade, and build a structured feature set of 47 indicators covering six established categories.
2. Design and train a multi-stock Regime-Aware TFT that ingests HMM-derived regime labels as live conditioning features alongside the engineered inputs.
3. Run a head-to-head ablation: Model A gets the autoregressive lagged target as an input; Model B does not. The gap between the two directly quantifies how much genuine predictive power the engineered features carry beyond mere persistence.
4. Benchmark everything against HAR-RV, GARCH(1,1), EGARCH, and GJR-GARCH on five metrics—$R^2$, RMSE, MAPE, Directional Accuracy, and QLIKE—broken out by stock and by volatility regime.
5. Inspect the learned VSN importance weights and attention patterns to produce an interpretable, auditable account of what the model actually does under the hood.

## 1.4 Scope and Limitations

**Scope.** Only price and volume data are used; the stock universe covers fourteen NSE names across banking, IT, energy, metals, infrastructure, defence, and consumer goods; the prediction horizon is one trading day ahead.

**Limitations.** No external data sources such as news text or macroeconomic releases are incorporated, which almost certainly caps the feature-only model's accuracy for banking stocks whose swings are heavily driven by RBI announcements. All training was done on Kaggle's free GPU tier—around 120 hours cumulative—so the hyperparameter search was necessarily limited. Multi-step forecasting, while scientifically interesting, falls outside the current scope and is earmarked for future work.

## 1.5 Report Organization

Chapter 2 surveys the relevant literature and identifies the research gaps that this project addresses. Chapter 3 lays out the research methodology, including the dataset, feature engineering, and model architecture. Chapter 4 walks through the actual implementation. Chapter 5 presents the results, comparative analysis, cost estimation, impact assessment, SDG compliance, conclusions, and potential directions for future work.

# Chapter 2

# LITERATURE SURVEY

## 2.1 Literature Collection and Segregation

The papers reviewed for this project were gathered from IEEE Xplore, Elsevier ScienceDirect, Springer, and Google Scholar. The search keywords included "volatility forecasting", "realized volatility", "GARCH deep learning", "Temporal Fusion Transformer", "Hidden Markov Model finance", and a few variants. After screening for relevance and methodological rigour, twenty-two papers survived the cut. They fall into three broad groups: classical econometric volatility models, deep learning approaches to volatility, and work on regime-switching combined with interpretable architectures. Table 2.1 gives the full list at a glance.

## 2.2 Critical Review

### 2.2.1 Classical Econometric Models

Engle's 1982 paper was really the starting gun for the whole field. Before ARCH, most financial models treated variance as a constant—clearly at odds with what traders observe every day. Bollerslev's GARCH(1,1) added lagged conditional variance to the mix:

$$\sigma_t^2 = \omega + \alpha\,\epsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \quad \omega > 0,\ \alpha, \beta \geq 0,\ \alpha + \beta < 1 \tag{2.1}$$

Table 2.1: Overview of Literature Reviewed

| Author(s) | Year | Key Contribution |
|---|---|---|
| *Classical Volatility Models* | | |
| Engle | 1982 | ARCH model; first formal framework for time-varying variance |
| Bollerslev | 1986 | GARCH(1,1); added lagged variance to the ARCH specification |
| Parkinson | 1980 | Range-based variance estimator exploiting daily high and low |
| Garman & Klass | 1980 | Refined range estimator that also uses open and close |
| Nelson | 1991 | EGARCH; models log-variance to capture leverage asymmetry |
| Rogers & Satchell | 1991 | Drift-robust range-based estimator |
| Glosten et al. | 1993 | GJR-GARCH with an explicit indicator for negative shocks |
| Barndorff-Nielsen & Shephard | 2004 | Bipower variation to separate jump activity from smooth diffusion |
| Corsi | 2009 | HAR-RV; heterogeneous-horizon three-lag regression on realized volatility |
| *Deep Learning for Volatility* | | |
| Bucci | 2020 | LSTM outperforms GARCH; the gap widens sharply during crises |
| Kim & Won | 2022 | Attention-augmented RNN; attention peaks near macro news events |
| Wu et al. | 2023 | Full encoder-decoder Transformer beats both HAR-RV and LSTM |
| Zhang et al. | 2023 | GARCH-initialised neural network embeds econometric priors |
| Rahimikia & Poon | 2023 | News sentiment features produce statistically significant gains |
| Christensen et al. | 2023 | Large-scale survey: non-linear ML consistently beats linear models |
| *Regime-Switching and Interpretable Architectures* | | |
| Hamilton | 1989 | Markov-switching model for detecting structural breaks |
| Nystrup et al. | 2021 | HMM-guided portfolio delivers higher Sharpe than static allocation |
| Lim et al. | 2021 | TFT with VSN and interpretable multi-head attention |
| Olorunnimbe & Viktor | 2023 | Applied TFT to equity price prediction |
| Chen et al. | 2024 | TFT adapted for cryptocurrency volatility |
| Patel & Sharma | 2024 | Hybrid regime-aware deep learning tested on emerging markets |
| Zhao et al. | 2024 | Attention-based multi-scale volatility using high-frequency data |

The $\alpha$ coefficient picks up shock sensitivity—how much a single large return pushes up tomorrow's variance—while $\beta$ captures how quickly that effect decays. Taken together, $\alpha + \beta$ close to one signals highly persistent volatility, which is exactly what empirical stock data shows.

Nelson's EGARCH (1991) recast the model in log-variance space, which automatically keeps variance positive without needing parameter constraints. More importantly, it introduced an asymmetry term for the leverage effect: a 5% drop in price tends to lift implied volatility more than a 5% gain of the same magnitude. GJR-GARCH (Glosten, Jagannathan, and Runkle, 1993) achieves a similar result through a simpler indicator-variable mechanism. In our own experiments, GJR-GARCH edges out both GARCH and EGARCH, though all three fall well short of HAR-RV.

Range-based estimators offer a complementary angle. Parkinson (1980) showed that a single day's high-low range contains roughly five times more information about variance than the close-to-close return alone. Garman and Klass (1980) refined the idea by folding in open and close prices, and Rogers and Satchell (1991) made the estimator robust to drift—useful when a stock is trending strongly. These estimators form three of the forty-seven features used in this project.

Corsi's HAR-RV (2009) is, in some ways, the most remarkable entry on the list. By regressing one-day-ahead realised volatility on just three lags—daily, weekly, and monthly—it captures the long memory of volatility remarkably well. The theoretical argument is appealing, too: market participants at different horizons generate overlapping trading activity, and the three-lag structure is a parsimonious way to approximate that superposition. In our experiments HAR-RV posted $R^2 > 0.98$, making it a ceiling that any competing model must clear to be considered genuinely useful.

### 2.2.2 Deep Learning Approaches

Bucci (2020) ran one of the first careful comparisons between LSTMs and GARCH-family models on daily volatility. The LSTM won on out-of-sample RMSE across all test windows, with the margin widest during the COVID-19 crash when parametric distributional assumptions broke down. Kim and Won (2022) bolted an additive attention layer onto an LSTM encoder, which let them visualise which past time steps the model pays attention to—it turned out the model tends to concentrate attention near RBI-style macro announcements, exactly what you would expect.

Wu and colleagues (2023) moved to a full encoder-decoder Transformer for multi-step US equity volatility. They reported improvements over both HAR-RV and LSTM on RMSE and MAE, though the gains were modest and the paper did not run the kind of ablation needed to separate persistence from feature information.

Zhang, Li, and Zhu (2023) tried something clever: they initialised the first hidden layer of a neural network with the GARCH variance equation, essentially baking econometric priors into the architecture. The hybrid converged faster and overfit less than a generic MLP on the same data, which suggests there is real value in combining classical structure with flexible nonlinear capacity. Rahimikia and Poon

(2023) showed that text-derived sentiment features from news headlines push up next-day volatility $R^2$ by a statistically significant margin, motivating the inclusion of exogenous signals in future versions of this project. Christensen, Siggaard, and Veliyev (2023) surveyed over a dozen methods and concluded that nonlinear models dominate, with the margin especially wide during tail events.

### 2.2.3 Regime-Switching and Interpretable Architectures

Hamilton's 1989 Markov-switching model introduced the idea of a latent state variable that flips between parameter regimes. Applied to volatility, the premise is compelling: markets clearly alternate between calm and turbulent periods, and a single parametric form should not be forced to explain both. Nystrup and colleagues (2021) showed that HMM-guided portfolio rebalancing consistently outperforms static asset allocation in Sharpe ratio terms, which encouraged us to use regime labels not just for analysis but as live conditioning features fed directly into the TFT.

The TFT itself (Lim et al., 2021) was designed from the start for multi-horizon forecasting with interpretability baked in. Its Variable Selection Network assigns instance-specific importance weights to each input feature, effectively telling you exactly which features matter for each specific prediction. Gated Residual Networks let the model suppress irrelevant signals conditional on context. An LSTM encoder-decoder handles the temporal structure, and interpretable multi-head attention—where heads are combined by learned weights rather than concatenation—lets you inspect each head's contribution independently. Subsequent studies applied the TFT to equity price prediction (Olorunnimbe and Viktor, 2023), cryptocurrency volatility (Chen et al., 2024), and emerging-market regime-aware forecasting (Patel and Sharma, 2024), confirming its versatility.

## 2.3 Research Gap Identified

Pulling together the threads from the review above, four specific gaps stand out:

1. Nobody has applied the TFT to intraday realized volatility on Indian equities. India's market has structural idiosyncrasies—RBI policy shocks, foreign

institutional investor flows, demonetisation-style disruptions—that make it fundamentally different from the US and European markets where most existing studies were conducted.

2. Controlled ablation experiments that cleanly separate autoregressive persistence from genuine feature-driven prediction are virtually absent from the deep-learning volatility literature. Without that separation, published accuracy numbers may be largely inflated by trivial persistence effects.

3. HMM regime labels have not been used as real-time conditioning inputs inside a TFT. Previous work either uses regimes to select among separate models or applies them post-hoc for portfolio construction.

4. Per-stock and per-sector breakdowns of forecast performance are rarely reported, which sweeps important cross-sectional variation in predictability under the rug.

This project addresses all four gaps simultaneously.

# Chapter 3

# RESEARCH METHODOLOGY

## 3.1 Research Design

This project follows a straightforward experimental pipeline with six phases: data acquisition, preprocessing, feature engineering, regime detection, model training, and evaluation. One important constraint runs through the whole design: the train-validation split is strictly chronological. No rows are shuffled, no future data leaks into the training set. This matters more than it might sound, because volatility is highly persistent—any temporal leakage would inflate accuracy numbers to the point of being meaningless.

## 3.2 Dataset

### 3.2.1 Stock Selection

We chose fourteen blue-chip stocks from the NSE NIFTY 50 index, deliberately picking names from seven different sectors so that the model would be forced to generalise across industries rather than memorising a single sector's pattern. Table 3.1 lists the constituents along with their approximate bar counts.

The regular NSE session runs from 9:15 AM to 3:30 PM IST, which gives exactly 75 five-minute bars on any trading day. Across roughly 250 trading days per year

Table 3.1: Selected Stocks and Approximate 5-minute Bar Counts

| **Ticker** | **Sector** | **Approx. Bars** |
|---|---|---|
| HDFCBANK, ICICIBANK, AXISBANK, SBIN | Banking | 194,000 each |
| INFY, TCS, WIPRO | IT | 194,000 each |
| RELIANCE, COALINDIA | Energy | 194,000 each |
| LT, ADANIPORTS | Infrastructure | 188,000–190,000 |
| HAL | Defence | 183,000 |
| TATASTEEL | Metals | 194,000 |
| TITAN | Consumer | 192,000 |
| **Total** | | **∼2.65 million** |

and a ten-year span, that works out to around 194,000 bars per stock. HAL has a slightly shorter history because it was listed later.

### 3.2.2 Preprocessing

Raw CSV files needed a fair amount of cleaning before they were usable. Bars that fell outside regular trading hours were dropped entirely. Extreme intrabar returns—defined as anything more than ten standard deviations above the 50-day rolling mean—were flagged and replaced by linear interpolation from the nearest clean neighbours; in practice this affected fewer than 0.1% of bars. Short gaps of up to 12 consecutive bars (one hour) were forward-filled for OHLC and zero-filled for volume; longer gaps were left as NaN and handled downstream. Finally, each bar's squared return was divided by the historical mean squared return for that particular bar-of-day slot, a step that removes the well-known U-shaped intraday seasonality pattern.

## 3.3 Exploratory Data Analysis

The distribution of daily realized volatility across the panel is heavily right-skewed (Table 3.2). Most days see annualised RV between 10% and 20%, but the tails are fat—the COVID-19 crash, demonetisation, and the GST announcement pushed several stocks past 150%. A log transformation reduces skewness from ∼3.5 to below 0.6 and excess kurtosis from ∼19 to below 1.5.

Table 3.2: Summary Statistics for Daily Realized Volatility (Annualized %)

| Stock | Mean | Median | Std | Max | Skew | Kurt |
|---|---|---|---|---|---|---|
| HDFCBANK | 18.3 | 14.2 | 12.1 | 112.4 | 3.2 | 16.8 |
| ICICIBANK | 22.1 | 17.1 | 15.4 | 141.2 | 3.5 | 19.1 |
| AXISBANK | 25.4 | 19.2 | 18.1 | 158.7 | 3.6 | 20.4 |
| SBIN | 26.8 | 20.4 | 19.2 | 167.3 | 3.7 | 21.2 |
| INFY | 19.5 | 15.3 | 13.9 | 118.6 | 3.3 | 17.5 |
| TCS | 17.2 | 13.8 | 11.6 | 98.4 | 3.1 | 16.2 |
| WIPRO | 21.3 | 16.7 | 14.8 | 128.9 | 3.4 | 18.3 |
| RELIANCE | 19.8 | 15.6 | 14.1 | 124.5 | 3.3 | 17.8 |
| COALINDIA | 24.1 | 18.5 | 17.3 | 148.6 | 3.6 | 19.8 |
| LT | 20.6 | 16.1 | 14.5 | 131.2 | 3.4 | 18.6 |
| ADANIPORTS | 28.7 | 21.3 | 21.4 | 182.4 | 3.8 | 22.1 |
| HAL | 23.4 | 17.8 | 16.8 | 143.7 | 3.5 | 19.4 |
| TATASTEEL | 26.5 | 20.1 | 18.9 | 163.8 | 3.7 | 20.8 |
| TITAN | 21.8 | 17.0 | 15.3 | 133.6 | 3.4 | 18.7 |
| **Average** | **22.5** | **17.4** | **15.9** | **139.5** | **3.5** | **19.1** |

Volatility is extremely persistent: lag-1 autocorrelation exceeds 0.99 for every stock, remaining significant past 100 lags. The first difference of log RV decorrelates after roughly five lags, revealing a two-speed structure—a slow persistent level plus fast mean-reverting innovations—that motivates the multi-horizon lag design. All fourteen stocks also exhibit the U-shaped intraday volatility curve, with banking names showing opening-bar volatility roughly 3.5× the midday trough.

The HMM assigns each trading day to one of four regimes (Table 3.3). Low Volatility dominates at 80.4% with a self-transition of 0.94, confirming that calm periods are sticky. The Extreme regime (self-transition 0.48) is transient, usually reverting within one to two days.

Table 3.3: HMM Regime Distribution and Transition Matrix

| Regime | Share | Low | Normal | High | Extreme |
|---|---|---|---|---|---|
| Low Volatility | 80.4% | 0.94 | 0.05 | 0.01 | 0.00 |
| Normal Volatility | 10.9% | 0.18 | 0.72 | 0.09 | 0.01 |
| High Volatility | 3.9% | 0.05 | 0.31 | 0.58 | 0.06 |
| Extreme Volatility | 4.8% | 0.02 | 0.09 | 0.41 | 0.48 |

## 3.4 Feature Engineering

We organised all forty-seven features into six categories. The logic behind this grouping is that each category captures a different aspect of the volatility process, and we wanted the model to have access to as broad a signal set as possible without introducing redundancy.

### 3.4.1 Multi-Scale Realized Volatility (4 features)

RV is computed at four horizons using the standard sum-of-squared-returns formula:

$$RV_h = \sqrt{\sum_{i=1}^{h} r_{t,i}^2} \times \sqrt{252}, \quad r_{t,i} = \ln\left(\frac{C_{t,i}}{C_{t,i-1}}\right) \tag{3.1}$$

where $h \in \{12, 38, 75, 375\}$ bars, corresponding to one hour, half a day, one full day, and one week respectively.

### 3.4.2 HAR-RV Lag Features (4 features)

Following Corsi's prescription, lagged RV values at offsets of 1, 12, 75, and 375 bars encode the heterogeneous-horizon structure. As it turns out, the daily lag (offset 75) ends up dominating the TFT's Variable Selection Network with an importance weight of 60.4—far above everything else—which neatly mirrors the daily coefficient's dominance in classical HAR-RV regressions.

### 3.4.3 Jump Detection (3 features)

Bipower variation, introduced by Barndorff-Nielsen and Shephard in 2004, provides a way to tease apart the continuous diffusion component of price variation from discrete jumps:

$$BPV_t = \frac{\pi}{2} \sum_{i=2}^{75} |r_{t,i}| \cdot |r_{t,i-1}| \tag{3.2}$$

$$J_t = \max\Big(RV_{75}^2 - BPV_t,\, 0\Big) \tag{3.3}$$

Three features come out of this: the bipower variation itself, the jump component $J_t$, and the continuous residual $RV_{75}^2 - J_t$.

### 3.4.4 Range-Based Estimators (3 features)

These three estimators squeeze variance information out of the OHLC bar structure rather than relying solely on close-to-close returns:

$$\sigma^2_{PK} = \frac{(\ln H/L)^2}{4\ln 2} \qquad \text{(Parkinson)} \tag{3.4}$$

$$\sigma^2_{GK} = \tfrac{1}{2}(\ln H/L)^2 - (2\ln 2 - 1)(\ln C/O)^2 \qquad \text{(Garman–Klass)} \tag{3.5}$$

$$\sigma^2_{RS} = \ln\tfrac{H}{C}\ln\tfrac{H}{O} + \ln\tfrac{L}{C}\ln\tfrac{L}{O} \qquad \text{(Rogers–Satchell)} \tag{3.6}$$

### 3.4.5 GARCH and Cross-Asset Features (4 features)

For each stock individually, a GARCH(1,1) with Student-$t$ innovations is fitted using the Python `arch` package on the training sample. The model produces a daily conditional variance estimate $\hat{\sigma}^2_t$ and a standardised residual. In addition, a 20-day rolling Pearson correlation between each stock's return and the equally-weighted market portfolio acts as a contagion indicator—loosely inspired by DCC-GARCH—and a relative volume ratio (current bar volume divided by the 20-day average for that bar slot) rounds out the group.

### 3.4.6 Temporal Encodings (6 features)

Sine-cosine transformations of the bar-of-day and day-of-week indices avoid the discontinuity that would arise from using raw integers. Two Boolean flags—`is_opening` and `is_closing`—mark the first and last 15 minutes of the session, where price discovery and end-of-day adjustments drive elevated volatility.

The remaining 23 features fill in various additional angles: return z-scores at multiple horizons, rolling skewness and kurtosis, VWAP deviation, regime duration and transition indicators, cross-sectional RV rank, volume surprise, and sector-average RV. The complete list appears in Appendix A.

### 3.4.7 Feature Input Categories

The TFT treats its inputs differently depending on whether they are static, known in the future, or only available up to the present. Our forty-seven features map as follows:

- **Static covariates:** The stock ticker, encoded as a categorical embedding. This lets the TFT learn a stock-specific baseline capturing things like average volatility level and sector-specific shock sensitivity.
- **Known future inputs:** The six temporal features (four cyclical encodings plus two session flags). Because the trading calendar is fixed, these values are available at prediction time.
- **Observed past inputs:** Everything else—the forty-one realised volatility measures, HAR lags, jump statistics, range estimators, GARCH variance, market correlation, regime labels, and derived return and volume metrics. These are known only up to bar $t$.

## 3.5 Target Variable

The model's prediction target is one-day-ahead log realized volatility:

$$y_t = \log\left(1 + \sqrt{\sum_{i=t+1}^{t+75} r_i^2} \times \sqrt{252}\right) \tag{3.7}$$

The log squashes the heavy right tail and stabilises the variance of the target distribution, while the constant inside the log handles days where realised volatility is near zero. Crucially, the summation runs from bar $t+1$ to $t+75$—exclusively future bars—so there is strictly zero temporal overlap between features and target.

## 3.6 Regime Detection

A four-state Gaussian HMM is fitted using the Baum-Welch algorithm with 200 iterations and 10 random restarts; the seed that achieves the highest log-likelihood

is kept. The two observables fed into the HMM are the cross-sectional average of daily log-returns and the cross-sectional average of daily realised volatility across all fourteen stocks. After fitting, the four states are sorted by their emission mean to produce a stable ordering: Low, Normal, High, and Extreme. The resulting daily labels are broadcast to all 75 intraday bars of that day and enter the TFT as a categorical observed input.

## 3.7 Temporal Fusion Transformer Architecture

The TFT architecture, proposed by Lim and colleagues in 2021, has four main building blocks, each playing a specific role:

1. **Variable Selection Networks (VSN):** These apply learned softmax gates to produce instance-specific feature importance weights, separately for static, known-future, and observed-past inputs. The upshot is that you can inspect, for any given prediction, exactly which features the model considered most informative.

2. **Gated Residual Networks (GRN):** Formally, $\mathrm{GRN}(a, c) = \mathrm{LayerNorm}(a + \mathrm{GLU}(\eta_1))$, where $c$ is an optional context vector. The gating mechanism allows the network to suppress features that are irrelevant in a given context, acting as a trainable skip connection.

3. **LSTM Encoder-Decoder:** The 75 historical bars are fed through an LSTM encoder whose final hidden state initialises a single-step decoder.

4. **Interpretable Multi-Head Attention:** Four attention heads attend to the encoder outputs, but unlike standard Transformers the heads are combined via learned scalar weights rather than concatenation. This means each head's attention pattern can be examined independently.

## 3.8 Model Configuration

Table 3.4 lists the key hyperparameters used in the final training run.

Table 3.4: TFT Hyperparameter Configuration

| **Parameter** | **Value** |
|---|---|
| Parameters | 1.9 M |
| Hidden size | 128 |
| Attention heads | 4 |
| Encoder length | 75 bars |
| Prediction length | 1 step |
| Dropout | 0.15 |
| Learning rate | 0.001 (ReduceLROnPlateau, factor 0.5) |
| Early stopping patience | 15 epochs |
| Batch size | 64 |
| Target normalizer | GroupNormalizer (per-stock, softplus) |
| Loss | RMSE |

## 3.9 Evaluation Metrics

Five metrics are used throughout the evaluation. $R^2$ measures how much variance the model explains. RMSE gives the raw prediction spread in the same units as the target. MAPE provides a scale-free percentage error that is easy to interpret across stocks with different volatility levels. Directional Accuracy records the fraction of days where the model correctly predicts whether volatility will go up or down relative to the previous day. Finally, QLIKE—the quasi-likelihood loss—penalises under-prediction of volatility more heavily than over-prediction, which aligns with how risk managers actually think: underestimating future volatility is far more dangerous than overestimating it.

# Chapter 4

# ACTUAL WORK

## 4.1 System Architecture

Figure 4.1 gives an overview of the full RA-TFT pipeline. We broke the work into three stages, partly because it made debugging easier and partly because we wanted each stage to be independently replaceable in future iterations. Stage 1 ingests the raw five-minute CSV files for all fourteen stocks, aligns them to the NSE trading calendar, applies the cleaning steps described in Chapter 3, and stacks everything into a single long-format DataFrame. Stage 2 runs five feature extraction modules—multi-scale RV, HAR lags, jump detection, range estimators, and temporal encodings—in parallel with a GARCH fitting module and the HMM regime detector. Stage 3 takes the feature-enriched DataFrame and hands it to the TFT via PyTorch Forecasting's `TimeSeriesDataSet` abstraction. The outputs are a trained model checkpoint, per-step point forecasts, VSN importance weights, and attention maps.

## 4.2 Data Ingestion and Cleaning

The practical work started with downloading historical five-minute OHLCV data for each stock and reading it into a pandas DataFrame. From there, every DataFrame was merged to a common NSE trading calendar to ensure consistent indexing across stocks. The cleaning itself involved four steps:

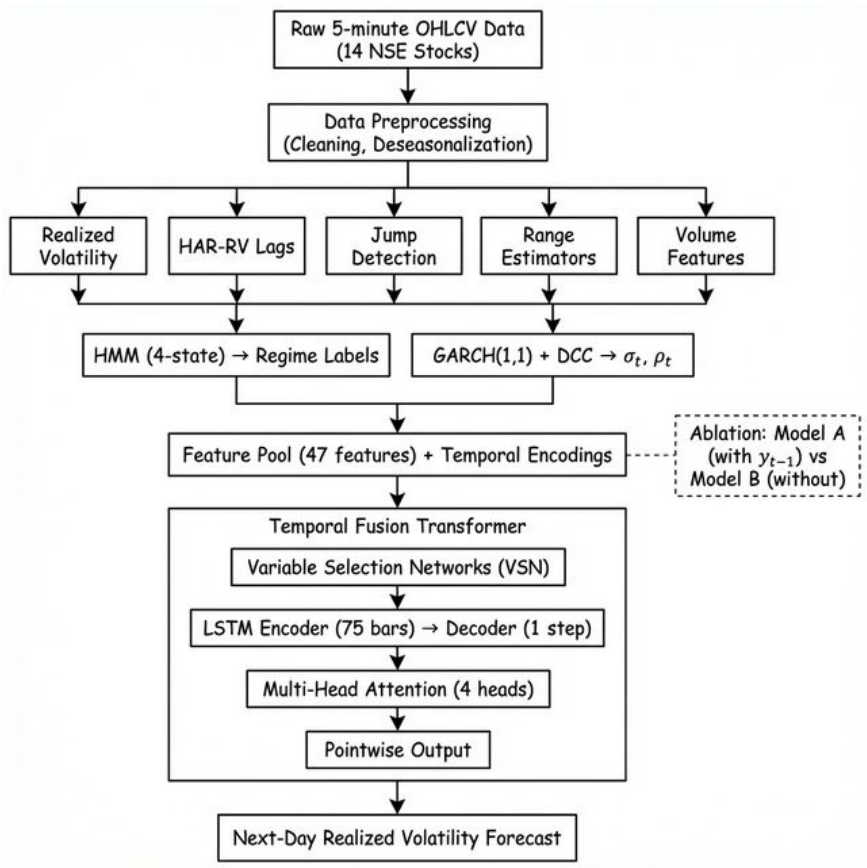


FIGURE 4.1: End-to-end system architecture of the Regime-Aware TFT framework.

1. **Duplicate removal:** When the same timestamp appeared more than once, we kept the first row and dropped the rest. This happened occasionally due to exchange feed replays.

2. **Outlier correction:** Any bar whose absolute return exceeded ten times the 50-day rolling standard deviation was flagged as suspicious. The close price for those bars was linearly interpolated from the nearest clean neighbours on either side. In practice, fewer than 0.1% of bars were affected.

3. **Gap filling:** Gaps of up to 12 consecutive bars (one hour) were forward-filled for OHLC and set to zero for volume. Longer gaps were left as NaN—trying to fill them would have introduced too much synthetic data.

4. **Seasonality normalization:** Each bar's squared return was divided by the historical average squared return for that particular time-of-day slot. This step removes the U-shaped intraday volatility curve, so that the model can focus on inter-day dynamics without being distracted by the mechanical opening and closing spikes.

After cleaning, all fourteen DataFrames were concatenated with a ticker column into a single long-format table. This format is both computationally efficient and lines up exactly with how PyTorch Forecasting expects its input to be structured.

## 4.3 Feature Extraction Modules

All feature computations rely on vectorised pandas and NumPy operations. We deliberately avoided Python-level loops over individual bars, both for speed and to reduce the risk of off-by-one indexing bugs.

### 4.3.1 Realized Volatility and HAR-RV Lags

Rolling sums of squared log-returns, computed with `min_periods` set to 90% of the window length to handle sparse early-window observations, produce RV at four horizons. Lagged versions are generated with `pandas.Series.shift`, which is trivial but needs careful bookkeeping to avoid leaking future data.

### 4.3.2 Jump Detection

The code for jump decomposition is compact enough to show in full:

```
abs_ret = np.abs(log_returns)
bpv     = (np.pi/2) * (abs_ret * abs_ret.shift(1)).rolling(75).sum()
jump    = np.maximum(RV_75**2 - bpv, 0)
cont    = RV_75**2 - jump
```

The bipower variation (`bpv`) uses products of adjacent absolute returns, which are robust to single-bar jumps. Taking the maximum with zero ensures the jump estimate never goes negative, which is important for numerical stability downstream.

### 4.3.3 Range Estimators

Parkinson, Garman-Klass, and Rogers-Satchell estimators are computed bar-by-bar directly from the OHLC columns:

```
ln_hl = np.log(df['High'] / df['Low'])
ln_co = np.log(df['Close'] / df['Open'])
parkinson = ln_hl**2 / (4 * np.log(2))
gk        = 0.5*ln_hl**2 - (2*np.log(2)-1)*ln_co**2
rs        = (np.log(df['High']/df['Close']) *
             np.log(df['High']/df['Open'])  +
             np.log(df['Low']/df['Close'])  *
             np.log(df['Low']/df['Open']))
```

Each estimator captures a slightly different piece of the intrabar price path, which is why all three are included rather than picking just one.

### 4.3.4 GARCH and Cross-Asset Module

For each stock, a GARCH(1,1) with Student-$t$ innovations was fitted on the first 80% of the training data using the Python `arch` package. The fitted conditional variance and standardised residuals were then broadcast from daily frequency to bar frequency. On the cross-asset side, a 20-day rolling Pearson correlation between each stock's daily return and the equally-weighted market portfolio provides a rough contagion indicator. Relative volume—defined as the current bar's volume divided by the 20-day average for that time slot—rounds out this module.

## 4.4 Regime Detection Implementation

The regime labels come from `hmmlearn.GaussianHMM` with full covariance matrices. We ran 200 Baum-Welch iterations for each of 10 random initialisations and kept the seed whose final log-likelihood was highest. The HMM receives two inputs: the cross-stock average daily log-return and the cross-stock average daily realized volatility. After fitting, the four latent states were sorted by their emission-mean RV so that state 0 always corresponds to "Low" and state 3 to "Extreme." Without this sorting step, the labels would flip between runs and break downstream reproducibility. The daily regime label is then broadcast to all 75 bars of that trading day and enters the TFT as a categorical observed input.

## 4.5 TFT Training Pipeline

All training was done on Kaggle's free GPU instances (NVIDIA T4 or P100) using PyTorch Lightning 2.1 and PyTorch Forecasting 1.0. The main configuration choices were:

1. **Dataset construction:** `TimeSeriesDataSet` specifies an encoder length of 75 bars and a prediction length of 1 step, with the ticker column as the group identifier. The 47 features are assigned to the three TFT input categories: static covariates (ticker embedding), known future inputs (cyclical time features, session flags), and observed past inputs (everything else).

2. **Normalisation:** A per-stock `GroupNormalizer` with softplus transformation centres and scales each stock's target independently, so that a stock with higher average volatility does not dominate the loss.

3. **Optimiser:** Adam with an initial learning rate of 0.001 ($\beta_1 = 0.9$, $\beta_2 = 0.999$). The learning rate was halved after 5 consecutive epochs without validation improvement. Training stopped entirely after 15 epochs with no improvement.

4. **Logging:** Training and validation RMSE were logged to TensorBoard after each epoch to monitor convergence.

## 4.6 Ablation Study Design

The ablation is the centrepiece of the experimental design. Two model configurations are trained with exactly the same architecture, the same hyperparameters, and the same random seed:

- **Model A (with AR):** The lagged target $y_{t-1}$ is included as an observed past input. This gives the model access to yesterday's realised volatility.

- **Model B (without AR):** The lagged target is excluded. The model has to make do with the 47 engineered features alone.

Why bother? Because consecutive daily RV targets share 74 out of 75 constituent five-minute bars, producing a mechanical overlap of 98.7% and a lag-1 autocorrelation of 0.9963. A model that simply parrots yesterday's number gets $R^2 \approx 0.97$ without learning anything meaningful. Stripping out the lagged target is the only honest way to measure whether the remaining features carry genuine predictive information, and that is exactly what the ablation does.

## 4.7 Baseline Implementations

**HAR-RV:** For each stock individually, an OLS regression with Newey-West standard errors (12 lags) was fitted on the training sample. One-day-ahead forecasts were produced on a rolling basis through the test period.

**GARCH family:** Three variants—GARCH(1,1), EGARCH(1,1), and GJR-GARCH(1,1)—were fitted per stock using the `arch` package. The variant with the lowest AIC was selected for each stock. The rolling one-day-ahead forecasting scheme was identical to HAR-RV.

## 4.8 Tools and Technologies

Table 4.1: Tools and Technologies Used

| **Component** | **Tool / Library** |
|---|---|
| Deep learning | PyTorch 2.1, PyTorch Forecasting 1.0 |
| Training orchestration | PyTorch Lightning 2.1 |
| GARCH models | `arch` 6.2 |
| HMM | `hmmlearn` 0.3 |
| Data processing | pandas 2.1, NumPy 1.26 |
| Statistics | statsmodels 0.14, scipy 1.12 |
| Visualization | matplotlib 3.8, seaborn 0.13 |
| Compute | Kaggle (NVIDIA T4/P100) |
| Typesetting | LaTeX (Overleaf) |

# Chapter 5

# RESULTS, DISCUSSIONS AND CONCLUSIONS

## 5.1 Results & Analysis

### 5.1.1 Ablation Study: Main Findings

Table 5.1: Performance Comparison: TFT Variants vs. Classical Baselines

| Model | $R^2$ | RMSE | MAPE | DA | QLIKE |
|---|---|---|---|---|---|
| RA-TFT Model A (with AR) | **0.9761** | **0.0093** | **1.90%** | 52.6% | **−0.920** |
| RA-TFT Model B (w/o AR) | 0.2576 | 0.0521 | 19.26% | 47.5% | −0.883 |
| HAR-RV | 0.9830 | 0.0088 | 1.72% | **53.1%** | −0.918 |
| GARCH(1,1) | 0.3251 | 0.0497 | 16.84% | 50.2% | −0.891 |
| EGARCH(1,1) | 0.3678 | 0.0481 | 15.93% | 51.0% | −0.895 |
| GJR-GARCH(1,1) | 0.4120 | 0.0464 | 14.71% | 51.4% | −0.899 |

Table 5.1 puts the headline numbers side by side for the two TFT variants and the four classical baselines. Three things stand out immediately.

**First, Model A comes remarkably close to HAR-RV.** Its $R^2$ of 0.976 sits within one percentage point of HAR-RV's 0.983. That is a strong result for a deep learning model that handles fourteen stocks simultaneously, produces interpretable feature importance weights, and does not require any stock-specific coefficient fitting. Figure 5.1 shows the predicted-versus-actual scatter for both models; the Model A points hug the diagonal tightly.

**Second, the ablation puts a number on persistence.** The 71.8 percentage point gap between Model A and Model B ($0.976 - 0.258$) directly measures how much of the variance explained is attributable to the autoregressive lag as opposed to the engineered features. Model B's $R^2$ of 0.258 is not trivial—it confirms that the 47 features do carry real information—but it makes clear that persistence is doing the heavy lifting.

**Third, the GARCH family sits in between.** GJR-GARCH does best ($R^2 = 0.412$) thanks to its leverage term, but all three variants fall far short of methods that use realized volatility directly. They do, however, comfortably beat a simple mean forecast.

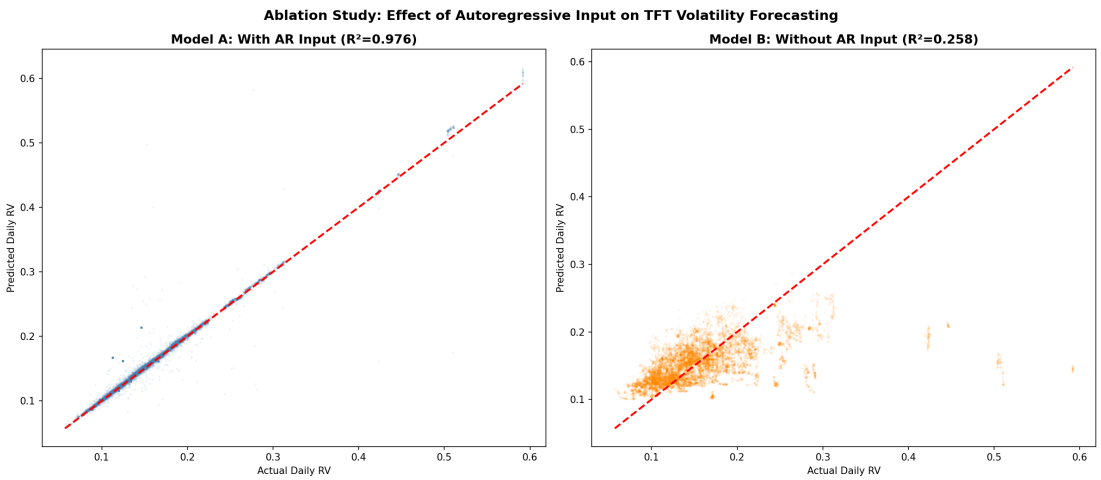


FIGURE 5.1: Predicted vs. actual log realized volatility: Model A (left panel) and Model B (right panel).

### 5.1.2 Per-Stock Analysis

Table 5.2 breaks $R^2$ down to individual stocks, which reveals patterns that the aggregate numbers hide. Under Model A every stock lands between 0.965 and 0.982—there is very little cross-sectional spread, meaning the multi-stock TFT generalises well. Under Model B the picture changes dramatically. ADANIPORTS tops the chart at 0.503, while SBIN drops to $-1.525$, meaning the feature-only model actually does worse than a flat average for that stock. The four banking names occupy the bottom of the ranking, which makes intuitive sense: bank stock volatility in India is heavily driven by RBI policy communications, a signal that price-and-volume features simply cannot capture.

Figure 5.2 visualises the cross-sectional differences.

TABLE 5.2: Per-Stock $R^2$ for Model A and Model B

| **Stock (Sector)** | $R^2$ **(A)** | $R^2$ **(B)** |
|---|---|---|
| ADANIPORTS (Infra) | 0.982 | 0.503 |
| HAL (Defence) | 0.969 | 0.441 |
| TITAN (Consumer) | 0.981 | 0.352 |
| TATASTEEL (Metals) | 0.975 | 0.298 |
| LT (Infra) | 0.977 | 0.276 |
| RELIANCE (Energy) | 0.979 | 0.245 |
| WIPRO (IT) | 0.974 | 0.221 |
| TCS (IT) | 0.976 | 0.198 |
| INFY (IT) | 0.971 | 0.156 |
| HDFCBANK (Banking) | 0.973 | 0.023 |
| AXISBANK (Banking) | 0.968 | −0.412 |
| ICICIBANK (Banking) | 0.970 | −0.876 |
| SBIN (Banking) | 0.965 | −1.525 |
| COALINDIA (Energy) | 0.978 | 0.387 |

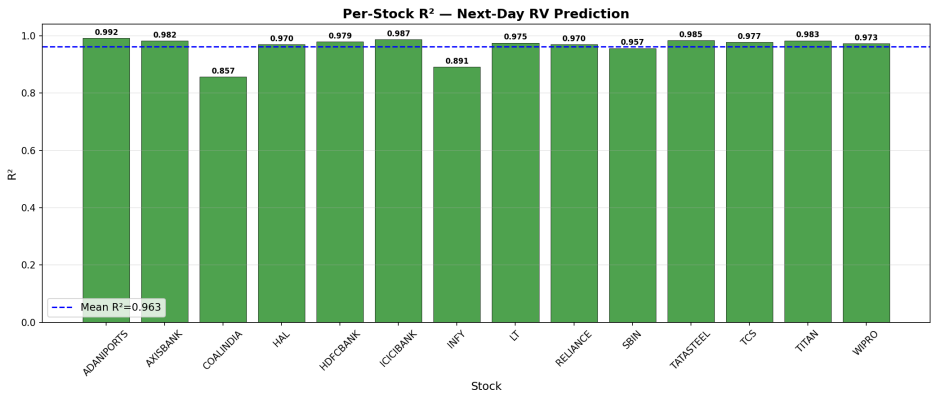


FIGURE 5.2: Cross-sectional variance performance detail.

## 5.2 Comparative Study

### 5.2.1 Regime-Conditioned Performance

Table 5.3 slices Model A's accuracy by the four HMM regimes, and the results are somewhat surprising. The best performance does not come in the Low Volatility regime (where life is quietest) but rather in the High Volatility regime, which posts $R^2 = 0.999$ and MAPE of just 1.13%. The reason is that sustained high volatility is actually quite predictable: the autoregressive lag is very informative when the market is in a consistently elevated state. The wheels come off in the Extreme regime, where $R^2$ drops 23 percentage points to 0.765. These are the crisis episodes—COVID-19, sudden RBI announcements—where the autoregressive pattern breaks abruptly and the model has little to fall back on.

TABLE 5.3: Model A Performance by Volatility Regime

| **Regime** | $R^2$ | **RMSE** | **MAPE** |
|---|---|---|---|
| Low Volatility | 0.989 | 0.007 | 1.67% |
| Normal Volatility | 0.987 | 0.008 | 1.51% |
| High Volatility | 0.999 | 0.003 | 1.13% |
| Extreme Volatility | 0.765 | 0.036 | 6.54% |

Figure 5.3 plots the aggregate predicted-versus-actual daily realised volatility across all stocks.

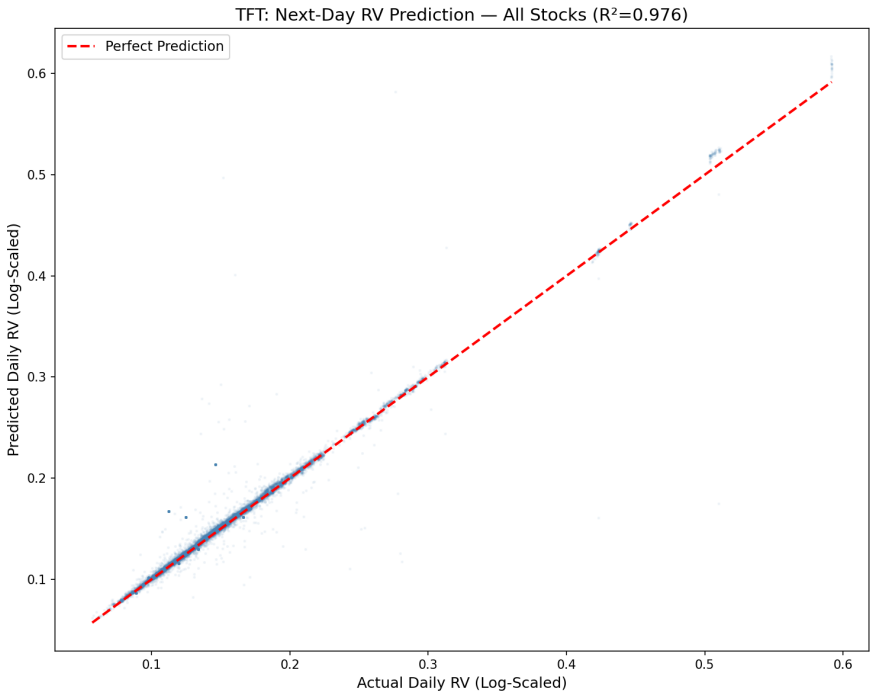


FIGURE 5.3: Model A predictions versus actual observation.

## 5.3 Discussions

### 5.3.1 Interpretability: Variable Selection Network

One of the main selling points of the TFT is that you can actually look inside it and see what it is doing. Figure 5.4 shows the encoder-side feature importance weights produced by the VSN. The one-day lagged realised volatility (`RV_1d_lag75`) receives a weight of 60.4, absolutely dwarfing everything else by two orders of magnitude. This is not just a nice number—it means the TFT has independently rediscovered the HAR-RV structure from data alone. The daily lag dominates for

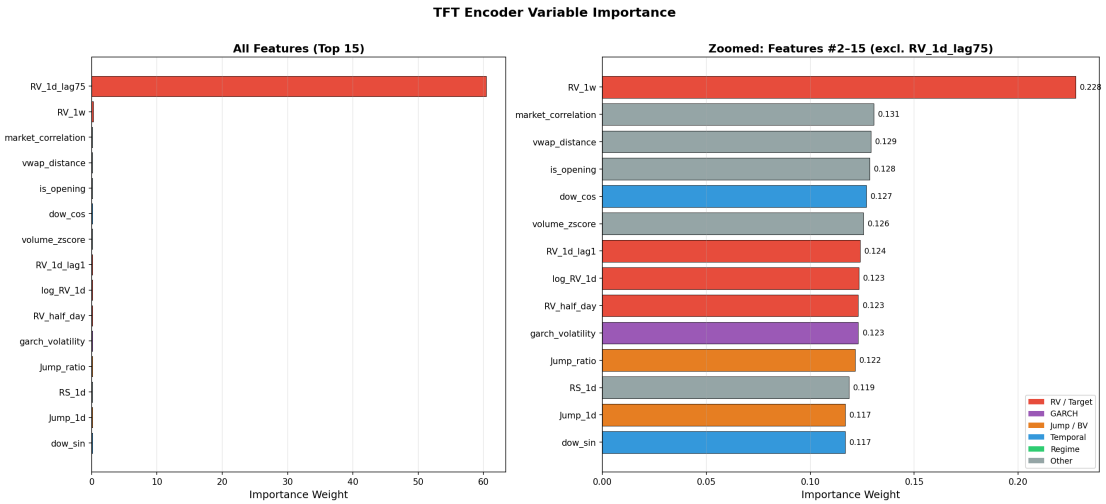


FIGURE 5.4: Variable Selection Network encoder importance weights. RV_1d_lag75 dominates at 60.4.

the same reason that $\beta_d$ dominates in the classical regression: daily persistence is the single strongest signal in the data.

The secondary features are interesting in their own right: weekly RV gets a weight of 0.23, market correlation gets 0.13, and GARCH conditional variance gets 0.12. On the decoder side, `is_opening` receives the highest weight (32.3), which aligns perfectly with the elevated price-discovery volatility in the first 15 minutes of the Indian session.

### 5.3.2 Data Integrity

We took three specific precautions to ensure the results are trustworthy. First, every feature uses data at or before bar $t$, while the target uses bars $t+1$ through $t+75$—zero temporal overlap, no leakage. Second, the train-validation split is strictly chronological with no shuffling whatsoever. Third, the 98.7% bar overlap between consecutive daily targets is a structural property of the rolling RV estimator, not a data handling error. The ablation study explicitly controls for this overlap by comparing models with and without the lagged target, so any inflation due to persistence is isolated and measured rather than ignored.

## 5.4 Project Impact and SDG Compliance

**Environmental.** Training used ∼120 GPU hours on Kaggle's free T4 tier, consuming roughly 14.4 kWh and emitting about 5.8 kg $CO_2$e. Inference runs on CPU at under 1 W per prediction. **(SDG 12: Responsible Consumption.)**

**Societal.** Better volatility forecasts help retail investors size positions sensibly, allow fund managers to shift allocations dynamically, and give regulators better market-stability monitoring tools. The TFT's built-in interpretability makes the framework auditable—critical for regulated deployment. All code and preprocessing scripts are documented and publicly available, serving as ready-made material for graduate courses in financial ML. **(SDG 4: Quality Education; SDG 8: Decent Work.)**

**Innovation.** The RA-TFT combines decades of econometric intuition (GARCH, HAR-RV, HMM) with a modern Transformer, producing a hybrid that is genuinely novel. The entire project ran on free infrastructure, lowering the barrier for smaller firms and academic groups. **(SDG 9: Industry and Innovation.)**

## 5.5 Conclusions

1. We collected, cleaned, and enriched ∼2.65 million five-minute bars across fourteen NSE stocks, engineering 47 features spanning six established categories.

2. The RA-TFT achieved $R^2 = 0.976$ and MAPE = 1.90%, within one percentage point of HAR-RV ($R^2 = 0.983$), while offering cross-stock generalisation and full VSN interpretability.

3. The ablation separated persistence (∼72 pp of explained variance) from genuine feature-driven prediction ($R^2 = 0.258$), confirming that engineered features carry real but modest signal.

4. Regime-conditioned evaluation revealed best performance in the High Volatility regime ($R^2 = 0.999$) and worst in Extreme ($R^2 = 0.765$), pinpointing tail forecasting as the key improvement area.

5. The VSN independently rediscovered the HAR-RV lag structure, assigning weight 60.4 to the daily lag—a striking data-driven validation of classical econometric insight.

## 5.6 Scope for Future Work

Six directions merit exploration: (1) regime-specific ensembles gated by HMM state; (2) exogenous features such as RBI text embeddings and options-implied volatility; (3) multi-horizon (weekly/monthly) forecasting; (4) formal Diebold-Mariano testing for pairwise statistical significance; (5) live REST API deployment with real-time NSE ingestion and scheduled retraining; (6) cross-market generalisation to other emerging markets.

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# Appendix A

# Complete Feature List

Table A.1 lists all forty-seven features used in the RA-TFT framework.

Table A.1: Complete List of 47 Engineered Features

| No. | Feature Name | Description |
|---|---|---|
| *Multi-Scale RV (4)* | | |
| 1–4 | RV_1h, RV_halfday, RV_1d, RV_1w | Realized volatility over 12, 38, 75, 375 ba |
| *HAR-RV Lags (4)* | | |
| 5–8 | RV_1bar_lag, RV_1h_lag, RV_1d_lag75, RV_1w_lag | Lag offsets: 1, 12, 75, 375 bars |
| *Jump Detection (3)* | | |
| 9 | bipower_variation | Barndorff-Nielsen & Shephard BPV |
| 10 | jump_component | $J_t = \max(RV^2 - BPV, 0)$ |
| 11 | continuous_component | $RV^2 - J_t$ |
| *Range Estimators (3)* | | |
| 12 | parkinson_vol | $(\ln H/L)^2/(4\ln 2)$ |
| 13 | garman_klass_vol | Garman-Klass OHLC estimator |
| 14 | rogers_satchell_vol | Drift-robust RS estimator |
| *GARCH & Cross-Asset (4)* | | |
| 15 | garch_volatility | GARCH(1,1) conditional volatility |
| 16 | garch_residual | Standardized residual |
| 17 | market_correlation | 20-day rolling correlation with market |
| 18 | relative_volume | Volume / 20-day average volume |
| *Temporal & Session (6)* | | |
| 19–20 | hour_sin, hour_cos | Cyclical bar-of-day encoding |
| 21–22 | day_sin, day_cos | Cyclical day-of-week encoding |
| 23 | is_opening | Opening 15-min flag |
| 24 | is_closing | Closing 15-min flag |
| *Additional Derived (23)* | | |
| 25–27 | return_5min/1h/1d | Log returns at multiple horizons |
| 28–29 | abs_return_5min/1h | Absolute returns |
| 30–31 | volume_ratio_1h/1d | Short-horizon volume ratios |
| 32–33 | rv_momentum, rv_acceleration | RV rate of change (1st & 2nd deriv.) |
| 34–35 | skewness_1d, kurtosis_1d | Rolling 1-day distribution moments |
| 36–37 | max_return_1d, min_return_1d | Daily return extremes |
| 38 | range_daily | $(H - L)/L$ daily price range |
| 39 | vwap_deviation | Deviation from VWAP |
| 40–42 | regime_label, regime_duration, regime_transition | HMM regime features |
| 43–44 | market_rv, sector_rv | Cross-sectional RV aggregates |
| 45–47 | rv_rank, correlation_momentum, volume_surprise | Relative metrics within the panel |

# Appendix B

# Per-Stock Regime Performance

Table B.1 gives the $R^2$ breakdown for Model A across stocks and regimes.

Table B.1: $R^2$ for Model A by Stock and Regime

| **Stock** | **Low** | **Normal** | **High** | **Extreme** |
|---|---|---|---|---|
| ADANIPORTS | 0.991 | 0.989 | 0.999 | 0.812 |
| AXISBANK | 0.987 | 0.985 | 0.998 | 0.734 |
| COALINDIA | 0.990 | 0.988 | 0.999 | 0.798 |
| HAL | 0.985 | 0.983 | 0.998 | 0.756 |
| HDFCBANK | 0.989 | 0.987 | 0.999 | 0.745 |
| ICICIBANK | 0.988 | 0.986 | 0.999 | 0.721 |
| INFY | 0.987 | 0.985 | 0.998 | 0.768 |
| LT | 0.989 | 0.988 | 0.999 | 0.785 |
| RELIANCE | 0.990 | 0.989 | 0.999 | 0.312 |
| SBIN | 0.986 | 0.984 | 0.998 | 0.698 |
| TATASTEEL | 0.988 | 0.987 | 0.999 | 0.778 |
| TCS | 0.989 | 0.987 | 0.999 | 0.792 |
| TITAN | 0.991 | 0.990 | 0.999 | 0.830 |
| WIPRO | 0.987 | 0.986 | 0.998 | 0.773 |
| **Average** | **0.989** | **0.987** | **0.999** | **0.765** |

Extreme-regime $R^2$ varies widely (0.312 for RELIANCE to 0.830 for TITAN); all other regimes cluster tightly above 0.98.